Probability on Trees and Networks: Exercise 5.57 and a separate Pemantle–Peres author proof core

Russell Lyons and Yuval Peres (book)
Robin Pemantle and Yuval Peres (separate paper)

Fixed sources and roles. The primary exercise is from Russell Lyons and Yuval Peres, *Probability on Trees and Networks*, **Version of 19 August 2026**, corrected and updated paperback edition: `https://rdlyons.pages.iu.edu/prbtree/book_pb.pdf`. Exercise 5.57 and its 5.55 setup are on physical page 190 (printed page 173); its complete Comment is on physical page 631 (printed page 614). The original book and separate `https://rdlyons.pages.iu.edu/prbtree/errata2.pdf` remain distinct sources.

The separate author text is Robin Pemantle and Yuval Peres, *On which graphs are all random walks in random environments transient?*, `https://www2.math.upenn.edu/~pemantle/papers/gRWRE.pdf`, linked on Pemantle's publication page under the 1996 publication in *Random Discrete Structures*, edited by D. Aldous and R. Pemantle, IMA volume 76, Springer, New York, pp. 207–211. The actual retained file is an **undated author-hosted manuscript**: 7 physical pages, consisting of an unnumbered summary cover followed by printed pages 1–6. The PDF creation date in 2003 is a digital-generation date, not an inferred manuscript revision date. This is not presented as the 1996 publisher-typeset pagination. The book and paper are two physical works with separately identified authors.

Editorial note. This private draft retains the original exercise, its complete Comment, and the paper's full Theorem (i)/(ii), printed proof, Remarks 1–2, acknowledgement and original references. Book background below is uncounted. The paper supplies a general network proof core; it does not mention Exercises 5.55/5.57 or print a particular transient tree with $\operatorname{br} T = 1$. The background bridge is recorded separately and is not an expanded standalone author solution of 5.57. No proof of Exercise 3.4 is supplied. Material and fidelity checks are recorded separately; this private source collection is not an adopted item.

Bridge scope (editorial source bookkeeping). The stretching interpretation uses positive integer edge lengths. In the paper's construction $\gamma_1 = 1$ and each later γ_k is free to be chosen arbitrarily large subject to its displayed lower bound. Restricting these free choices to positive integers is the standard specialization used for this interpretation; the paper itself does not call them integers. The book's Series Law identifies L unit resistors in a path with one resistor of resistance L. The distinction between a merely transient tree and the $\operatorname{br} T > 1$ class is located by the original Exercise 3.4, its hint, the spherical criterion and the growth statement reproduced below. These background roles are not a newly constructed example of T or a new proof. No finite-mean or positive-branching-number condition is added to Exercise 5.57.

The numerical and event expressions in the paper are retained literally: its induction prints $\gamma_k > 2N_{k-1}D_{k-1}\gamma_{k-1}$, while its later estimate prints $\mathbf{P}(B_k) \le N_k D_k \gamma_k/\gamma_{k+1} \le 2^{-k}$. The displayed unindexed $\cup$ in B_k is also retained. No coefficient, event or intended erratum has been inserted. Public redistribution permission for the edited book/paper selection has not been established.

Original book setup and primary question

Source: Exercise 5.55, physical page 190, printed page 173; uncounted setup.

5.55. Randomly stretch a tree T by adding vertices to ("subdividing") the edges to make the edge $e(x)$ a path of length $L_x(\omega)$ with L_x i.i.d. Call the resulting tree $T(\omega)$. Calculate the distribution of $\operatorname{br} T(\omega)$ in terms of $\operatorname{br} T$ and of the distribution of L_x. Show that if $\operatorname{br} T > 1$, then $\operatorname{br} T(\omega) > 1$ a.s.

Source: Exercise 5.57, physical page 190, printed page 173.

5.57. Consider the stretched tree $T(\omega)$ from Exercise 5.55. If we assume only that simple random walk on T is transient, must simple random walk on $T(\omega)$ be transient a.s.?

Source: Complete Comment 5.57, physical page 631, printed page 614.

5.57. See Pemantle and Peres (1996).

Uncounted same-book network and tree background

Source: Series Law and its printed proof, physical page 44, printed page 27.

I. Series Law. Two resistors[1] r_1 and r_2 in series are equivalent to a single resistor $r_1 + r_2$. In other words, if $w \in \mathsf{V}(G) \setminus (A \cup Z)$ is a node of degree 2 with neighbors u_1, u_2 and we replace the edges (u_i, w) by a single edge (u_1, u_2) having resistance $r(u_1, w) + r(w, u_2)$, then all potentials and currents in $G \setminus \{w\}$ are unchanged and the current that flows from u_1 to u_2 equals $i(u_1, w)$.

$$u_1 \quad — \quad w \quad — \quad u_2$$

Proof. It suffices to check that Ohm's and Kirchhoff's laws are satisfied on the new network for the voltages and currents given. This is easy. □

Source: Exercise 2.8, physical page 47, printed page 30; repeated on physical page 75, printed page 58.

Exercise 2.8. Let (G, c) be a network. A network automorphism of (G, c) is a map $\phi : G \to G$ that is a bijection of the vertex set with itself and a bijection of the edge set with itself such that if x and e are incident, then so are $\phi(x)$ and $\phi(e)$ and such that $c(e) = c(\phi(e))$ for all edges e. Suppose that (G, c) is spherically symmetric about o, meaning that if x and y are any two vertices at the same distance from o, then there is an automorphism of (G, c) that leaves o fixed and that takes x to y. Let C_n be the sum of $c(e)$ over all edges e with $d(e^-, o) = n - 1$ and $d(e^+, o) = n$. Show that

$$\mathcal{R}(o \leftrightarrow \infty) = \sum_{n \geq 1} \frac{1}{C_n},$$

whence the network random walk on G is transient iff

$$\sum_{n \geq 1} \frac{1}{C_n} < \infty.$$

Source: Section 3.3, physical page 99, printed page 82; lower-growth definition excerpt.

In this section, we again consider only locally finite, infinite trees. To understand better their branching number, we will look at the simpler notion of growth. For a vertex $x \in T$, let $|x|$ denote its distance from o. Define the lower (exponential) growth rate of a tree T by

$$\underline{\mathrm{gr}} T := \liminf_{n \to \infty} |T_n|^{1/n},$$

where $T_n := \{x \in T;\ |x| = n\}$ is level n of T.

Source: Section 3.3, physical page 100, printed page 83; selected paragraph.

We call T spherically symmetric if $\deg x$ depends only on $|x|$. Recall from Exercise 1.2 that $\mathrm{br}\, T = \underline{\mathrm{gr}} T$ when T is spherically symmetric.

Source: Exercise 3.4, physical page 99, printed page 82; collected again on physical page 109, printed page 92; Comment on physical page 622, printed page 605.

Exercise 3.4. For simple random walk on T to be transient, is it necessary that $\mathrm{br}\, T > 1$?
Comment 3.4. Consider spherically symmetric trees.

[1] A resistor r is an edge with resistance r. We have drawn such edges using the squiggly notation common to physics, but this only indicates that they have weights.

Separate author core: Pemantle and Peres

Source: Undated author copy, physical page 2, printed page 1; complete introductory mathematical text.

Simple random walks on infinite graphs have been widely studied with particular attention to the dichotomy between recurrence and transience. In the last two decades there has been interest in random walks in random environments (RWRE's), which are mixtures of Markov chains in which the transition probabilities are picked according to some prior distribution. For reversible Markov chains, the transition probabilities may be described by associating resistances $R(vw)$ to each edge (vw), the transition probabilities out of any vertex being proportional to the inverses of the resistances on the edges incident to that vertex:

$$p(v,w) = \frac{R(vw)^{-1}}{\sum_z R(vz)^{-1}} \tag{1}$$

where the sum is over all vertices z adjacent to v. One natural model[4], considered in Grimmett and Kesten (1984), involves taking the resistances as i.i.d. positive random variables. A result of Adams and Lyons (1991) implies that if a tree Γ has positive Hausdorff dimension, then for an arbitrary distribution of the resistances the resulting network is almost surely transient. (See Lyons (1990) where the term branching number is used for the exponential of the Hausdorff dimension.) The surprising aspect of this is that no conditions are imposed on the tail of the distribution. This note proves a converse of this implication, but on a general graph. Specifically, the implication $(a) \Rightarrow (b)$ in part (ii) of the theorem below shows that almost sure transience for an arbitrary resistance distribution on a tree Γ implies $p_c < 1$ for IID bond percolation. By results in Lyons (1990), this implies that Γ has positive dimension. Part (i) is immediate from the well known connections between random walks and electrical networks (c.f. Doyle and Snell (1984)) and is included for completeness.

Source: Complete Theorem, physical pages 2–3, printed pages 1–2.

Theorem: *Let G be a locally finite connected graph. Suppose that a positive random variable $R(e)$ is attached to every edge e of G so that the collection $\{R(e)\}$ is i.i.d.. Let μ denote their common distribution. Consider the RWRE given by (1).*

- (i) *If $\int x\, d\mu < \infty$ and simple random walk on G is transient, then the resistor network $\{R(e)\}$ has finite net resistance almost surely, thus defining almost surely a transient random walk.*
- (ii) *The following conditions on G are equivalent: (a) for any distribution μ on $(0,\infty)$ the resistor network $\{R(e)\}$ is almost surely transient; (b) there exists some $p < 1$ such that independent bond percolation on G with parameter p contains with positive probability a cluster on which simple random walk is transient.*

Source: Complete printed proof, physical pages 3–5, printed pages 2–4.

Proof. We rely on two facts from the theory relating reversible random walks to the corresponding resistor networks. Firstly, the random walk is transient if and only if the network has finite resistance from some (hence every) vertex to infinity, i.e., there is some nonzero flow $\{F(e)\}$, satisfying Kirchoff's laws except at one source vertex, such that the energy

$$\sum_e F(e)^2 R(e)$$

is finite. Secondly, Rayleigh's principle asserts that the resistance of a network is monotone in each of the individual resistances. Both facts may be found in Doyle and Snell (1984). The

[4] see Remark 1 for another RWRE model relevant to the present discussion

proof of (i) follows immediately: G is transient for simple random walk, so some flow $\{F(e)\}$ on G has finite energy

$$\sum_e F(e)^2$$

with respect to unit resistances; the expected energy of this flow in the network of random resistances is

$$\mathbf{E}\sum_e R(e)F(e)^2 = \int x\, d\mu \sum_e F(e)^2 < \infty$$

so the resistance is almost surely finite and hence the network is almost surely transient.

(ii) The direction $(b) \Rightarrow (a)$ is easy: assume (b) for some parameter p and consider the random subgraph of G consisting of those edges e for which $R(e) \le Q_p$, where $1 > \mu(0, Q_p) \ge p$. Resistances have the following linearity: multiplying every resistor by C multiplies the resistance of the network by C. Thus our assumption (b) implies there is a positive probability that some component of this subgraph will have finite resistances when all edges have resistance Q_p. By Rayleigh's law, decreasing the resistances on edges in this component from Q_p to $R(e)$ and decreasing all other resistances from ∞ to $R(e)$ gives a network with finite resistance, and therefore random walk on G with resistances $\{R(e)\}$ is transient with positive probability. Since this is a tail event for the countable collection $\{R(e)\}$, the probability is one.

Now we verify that $(a) \Rightarrow (b)$. Assume that for every $p < 1$, simple random walk on p-percolation clusters is almost surely recurrent. Let $p_k \uparrow 1$ and construct a probability measure μ on $(0, \infty)$ as follows. Fix a vertex $v \in G$ and set $\gamma_1 = 1$. Define a measure μ_1 on $\mathbf{R} \cup \{\infty\}$ by $\mu_1(\{\gamma_1\}) = p_1$ and $\mu_1(\{\infty\}) = 1 - p_1$. The RWRE on G with resistance distribution μ_1 is just simple random walk on the p_1-percolation clusters and is thus recurrent. Therefore there exists N_1 such that when choosing a random environment according to μ_1, the probability is at least $1/2$ that either all edges adjacent to v have infinite resistance (i.e. v is isolated) or else the resulting RWRE started from v returns to v in the first N_1 steps. Let D_1 be the maximal degree of any vertex in G within distance N_1 of v.

For the induction step, let $k \ge 2$ and assume that γ_j, μ_j, N_j and D_j have been defined for $1 \le j < k$ and $\mu_j(\infty) = 1 - p_j$. Pick $\gamma_k > 2N_{k-1}D_{k-1}\gamma_{k-1}$. Define μ_k by moving mass $p_k - p_{k-1}$ from infinity to γ_k, i.e.

$$\mu_k = \mu_{k-1} + (p_k - p_{k-1})(\delta_{\gamma_k} - \delta_\infty).$$

The RWRE with resistance distribution μ_k is recurrent since it yields an environment whose resistances on the p_k-percolation cluster are bounded between 1 and γ_k [use the aforementioned linearity together with Rayleigh's law]. Thus we may choose an integer N_k, such that if a random environment is chosen according to μ_k, the probability is at most 2^{-k} that v is not isolated and the resulting RWRE started from v fails to return to v within the first N_k steps. Finally, define D_k to be the maximal degree of any vertex of G within distance N_k of v.

Finally, set $\mu(\{\gamma_k\}) = p_k - p_{k-1}$ for all $k \ge 2$, with $\mu(\{\gamma_1\}) = p_1$, so that μ is the weak limit of the μ_k. We construct a μ-RWRE $\{S_n\}$ on the same probability space as a sequence $(\{S_n^{(k)}\} : k = 1, 2, 3, \ldots)$ of μ_k-RWRE's as follows. Let the resistances $\{R(e)\}$ be i.i.d. with common distribution μ and define

$$R^{(k)}(e) = \begin{cases} R(e) & \text{if } R(e) \le \gamma_k \\ \infty & \text{if } R(e) > \gamma_k. \end{cases}$$

Observe that this makes $\{R^{(k)}(e)\}$ i.i.d. with distribution μ_k. Let $\{S_n\}$ be a random walk starting from v with transition probabilities determined by the resistances $\{R(e)\}$. Note that if $R(wz) \le \gamma_k$ then the transition probability from w to z with resistances $\{R^{(k)}(e)\}$ is greater than or equal to the the transition probability from w to z with resistances $\{R(e)\}$. Thus for each k, we may define a random sequence $\{S_n^{(k)}\}$ so as to have transition probabilities determined by $\{R^{(k)}\}$ and so that $S_n^{(k)} = S_n$ for all $n \le T_k$, where T_k is the least time t for which $R(S_t S_{t+1}) > \gamma_k$.

Define events

$$\begin{aligned} A_k &= \{R(vw) > \gamma_k \text{ for all } w \text{ neighboring } v\} \subseteq \{T_k = 0\} \\ B_k &= \{R(S_j, S_{j+1}) \geq \gamma_{k+1} \text{ for some } 0 \leq j < N_k\} \setminus A_k \\ &= \cup\{0 \leq T_k < N_k\} \setminus A_k \\ C_k &= \{S_n^{(k)} \neq v \text{ for all } 1 \leq n \leq N_k\}. \end{aligned}$$

The event $G_k = \{S_n \neq v \text{ for all } 1 \leq n \leq N_k\}$ is contained in $A_k \cup B_k \cup C_k$, since on $G_k \setminus C_k$ the time T_k is less than N_k. For $1 \leq j < N_k$, the probability of the event $\{T_k = j\}$ is at most $D_k\gamma_k/\gamma_{k+1}$, since the sum of $R(e)^{-1}$ over edges incident to S_j with resistance greater than γ_k is at most D_k/γ_{k+1} and there is at least one edge incident to S_j with $R(e)^{-1} \geq \gamma_k^{-1}$. Similarly, $\mathbf{P}(\{T_k = 0\} \setminus A_k) \leq D_k\gamma_k/\gamma_{k+1}$. Thus $\mathbf{P}(B_k) \leq N_k D_k \gamma_k/\gamma_{k+1} \leq 2^{-k}$ by construction of γ_{k+1}. But N_k is defined so that $\mathbf{P}(C_k \setminus A_k) \leq 2^{-k}$ and clearly $\mathbf{P}(A_k) \leq 1 - p_k$. Therefore

$$\mathbf{P}(G_k) \leq (1 - p_k) + 2^{1-k} \longrightarrow 0$$

as $k \to \infty$ and the μ-RWRE is recurrent. □

Source: Remarks 1–2 and acknowledgement, physical page 6, printed page 5.

Remarks:

1. The recent paper of Grimmett, Kesten and Zhang (1991) shows that for Euclidean lattices $\mathbf{Z}^d$ ($d \geq 3$), random walk on supercritical percolation clusters is almost surely transient, so the theorem is applicable, and all RWRE's with i.i.d. positive resistances are transient. Note that the RWRE models discussed in Durrett (1986) and Sunyach (1987) may be recurrent even when $d \geq 3$. Resistances in Durrett's model are stationary under $\mathbf{Z}^d$-shifts, so his model is close to the present one.

2. By Lyons (1990), the critical probability for bond percolation on a tree Γ is $p_c(\Gamma) = e^{-\dim \Gamma}$ and if $\dim(\Gamma) > 0$ and $p > p_c$, then simple random walk on any infinite cluster is transient (since the cluster must have positive dimension). For a tree, property (b) is thus equivalent to $\dim(\Gamma) > 0$. The Max Flow Min Cut Theorem (as applied in Lyons 1990) allows us to formulate this purely in terms of flows:

   > On a tree, property (a) is equivalent to the existence of a flow $\{F(e)\}$ satisfying Kirchoff's laws (except at the root) for unit resistors, and having the exponential decay property:
   >
   > $$F(e) \leq C\rho^{\mathrm{dist}(e)}$$
   >
   > where $\mathrm{dist}(e)$ is the distance from e to the root of Γ.

Acknowledgement: Thanks to Harry Kesten for showing us a surprising example which led to this note.

Source: Original references, physical pages 6–7, printed pages 5–6.

References

[1] Adams, S. and Lyons, R. (1991). Amenability, Khazdan's property and percolation for trees, groups and equivalence relations. *Israel J. Math.* **75** 341–370.

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[4] Grimmett, G. and Kesten, H. (1984). Random electrical networks on complete graphs. *J. London Math. Soc.* **2** 171–192.

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[7] Sunyach, C. (1987). Sur la transience et la récurrence des marches aléatoires en milieu aléatoire. *Ann. Inst. H. Poin.* **23** 613–626.